Guided practice: prediction and variables

Circle one answer in each set of brackets; use boxes for multiple choices. Each written answer has two lines.

Information for this question

Bile (膽汁) emulsifies lipids (乳化脂質) into smaller droplets without changing the lipid molecules. Predict a visible change and link your reason to this action.

B1.1 Add 2 mL bile (膽汁) to the same oil–water mixture and shake it. Write one observable prediction and give your reason.

Prediction and reason:

Variable reminder

Example: change only the shaking method and observe the appearance. Shaking is the independent variable; appearance is the dependent variable. Keep other conditions the same.

B1.2 The main comparison changes the combination of digestive juices. Classify the factors as independent, dependent or controlled variables.

Independent variable (the factor changed on purpose)	Dependent variable (the result you measure or observe)	Controlled variables (the factors kept the same)
☐ Total volume of oil and water	☐ Total volume of oil and water	☐ Total volume of oil and water
☐ Reaction temperature	☐ Reaction temperature	☐ Reaction temperature
☐ Reaction time	☐ Reaction time	☐ Reaction time
☐ Total volume of liquid added	☐ Total volume of liquid added	☐ Total volume of liquid added
☐ Method of shaking	☐ Method of shaking	☐ Method of shaking
☐ Appearance of the mixture	☐ Appearance of the mixture	☐ Appearance of the mixture
☐ Combination of digestive juices	☐ Combination of digestive juices	☐ Combination of digestive juices

Conditions for a fair test

Change only the factor being tested and keep the other conditions equal. Some options below describe an unfair arrangement.

B1.3 Choose all the conditions needed for a fair comparison.

- ☐ Observe all setups after the same reaction time.
- ☐ The X + Y group can have more liquid in total to compare it with the other groups.
- ☐ Use the same starting amounts of oil and water and the same total volume of added liquid in each setup.
- ☐ Use the same method of shaking for every setup.

Guided practice: controls, labels and fairness

Keep starting oil/water amounts, total added volume, 37°C, observation time and shaking the same.

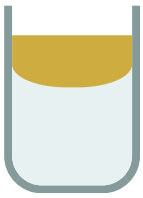
> Control group
> Compare the experimental group with a control group to judge the treatment’ s effect. Keep other conditions the same.

B2.1 To compare adding digestive juice with adding only water, which of the four sets is the control group? Why keep the total volume added the same?

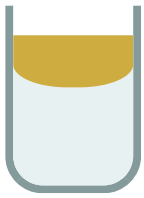
B2.2 Use the following recipes to complete the four setup labels and added volumes.

Recipes: water, 2 mL; X, 2 mL; Y, 2 mL; X + Y, 1 mL each. All start with the same amounts of oil and water.

Setup 1



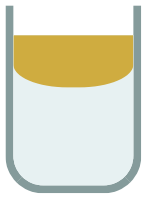
Setup 2



Setup 3



Setup 4



B2.3 Someone adds 2 mL water to the control but 2 mL X and 2 mL Y to the combined set, with other conditions equal. Is this fair? Identify the problem, fix the two volumes and explain why.

First find the total volume added to each set. Then check whether only one factor changed.

> Evidence reminder
> Appearance gives clues but cannot replace chemical product tests. Other tests and repeated tests are needed.

B2.4 Limits of appearance evidence: choose the best statement, then name another kind of evidence needed.

This experiment records the oil layer and cloudiness/clarity, without directly testing for digestion products. Which statement fits best? （Appearance provides clues about the effects of digestive juices, but alone cannot prove that glycerol (甘油) and fatty acids (脂肪酸) have formed. ／ A clear mixture completely proves that lipids have broken down into glycerol (甘油) and fatty acids (脂肪酸).）

Brief reason / explanation:

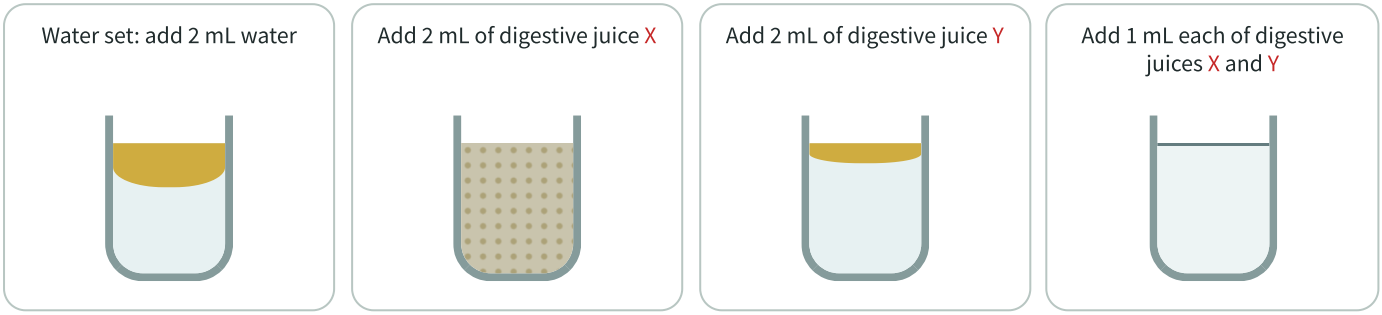
Complete and keep the first answers in B. After receiving C, correct in another colour without erasing the original.

Independent application: four main sets

Observation reminder
Look for layers, oil-layer thickness and cloudiness. Describe what you see before explaining why.

B3.1 Observe Figure 1 and record the four appearances.

Figure 1. Appearances in the four main sets



Setup / conditions	Your appearance record (one per set)	
Water set: add 2 mL water	☐ Separate oil and water layers (no clear change you can see) ☐ become cloudy	☐ Separate oil and water layers (thinner oil layer) ☐ become clear
Add 2 mL of digestive juice X	☐ Separate oil and water layers (no clear change you can see) ☐ become cloudy	☐ Separate oil and water layers (thinner oil layer) ☐ become clear
Add 2 mL of digestive juice Y	☐ Separate oil and water layers (no clear change you can see) ☐ become cloudy	☐ Separate oil and water layers (thinner oil layer) ☐ become clear
Add 1 mL each of digestive juices X and Y	☐ Separate oil and water layers (no clear change you can see) ☐ become cloudy	☐ Separate oil and water layers (thinner oil layer) ☐ become clear

Emulsification (乳化作用) reminder
Emulsification (乳化作用) makes smaller oil droplets without changing the lipid molecules.

B3.2 X: choose the appearance, then explain it using emulsification (乳化作用).

Digestive juice X mainly causes lipids to form （a clear solution ／ a cloudy emulsion (乳状液)）.

Brief reason / explanation:

Surface-area reminder
Spreading the same amount of lipid into smaller droplets lets lipase (脂肪酶) contact more surface. This is not a speed measurement.

B3.3 Total surface area after droplets become smaller: choose the answer and explain how this helps Y act.

Digestive juice X first breaks up large oil droplets into small droplets, causing their total surface area to （decrease ／ increase）, which helps digestive juice Y act.

Brief reason / explanation:

Complete and keep the first answers in B. After receiving C, correct in another colour without erasing the original.

Independent application: heating and evidence

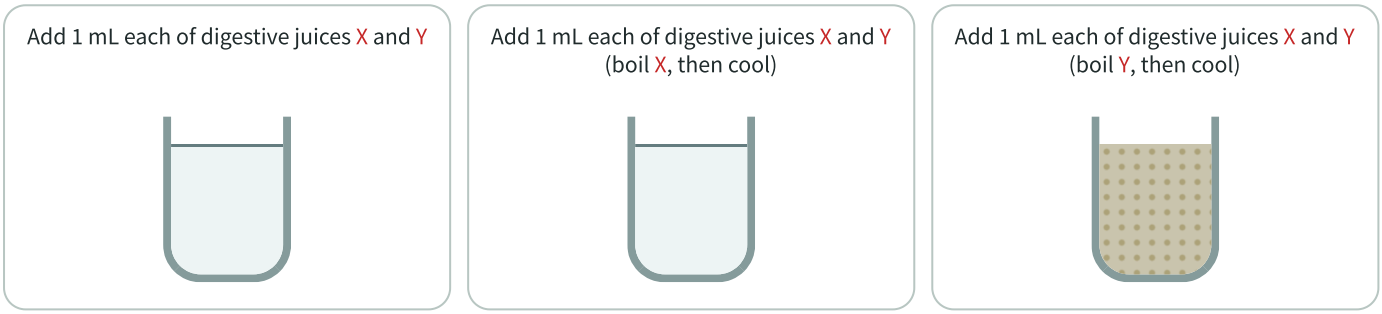
Information for the heating tests

Lipase (脂肪酶) is a protein. High heat can change its shape and stop it working (denaturation); cooling does not restore its action in this model. Cool boiled liquids to the common 37°C before a fair comparison.

B4.1 To study boiling one digestive juice, which set should be the control group? What must happen to the boiled liquid before comparison?

B4.2 Observe Figure 2, record the heated sets and compare them with the unboiled X + Y control group.

Figure 2. Comparison after boiling and cooling



Setup / conditions	Your appearance record (one per set)	
Add 1 mL each of digestive juices X and Y (boil X, then cool)	☐ become cloudy	☐ become clear
Add 1 mL each of digestive juices X and Y (boil Y, then cool)	☐ become cloudy	☐ become clear

B4.3 Juice that loses its action after heating: choose the answer and explain why cooling does not restore its action.

Extension conclusion: the digestive juice that loses its function after boiling is（Digestive juice X ／ Digestive juice Y）.

Brief reason / explanation:

Limits of these data

Lipase (脂肪酶) hydrolyses lipids into glycerol (甘油) and fatty acids (脂肪酸). No speed was measured. X + Y uses 1 mL each; single-juice sets use 2 mL. This cannot prove a measured combined effect or faster rate. Name the compared sets and differences.

B4.4 Use at least one main-set comparison and one heating comparison to explain emulsification (乳化作用), enzyme action and denaturation. Name the sets. Do not claim a measured combined effect or increase in speed.

Complete and keep the first answers in B. After receiving C, correct in another colour without erasing the original.